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Assignment 23: Network connectivity diagnosis and configuration

https://github.com/elovric13/eutech/tree/network-connectivity-diagnosis-and-configuration-elovric13

1. **Identify the active network connection on your computer (Wi-Fi or Ethernet).**

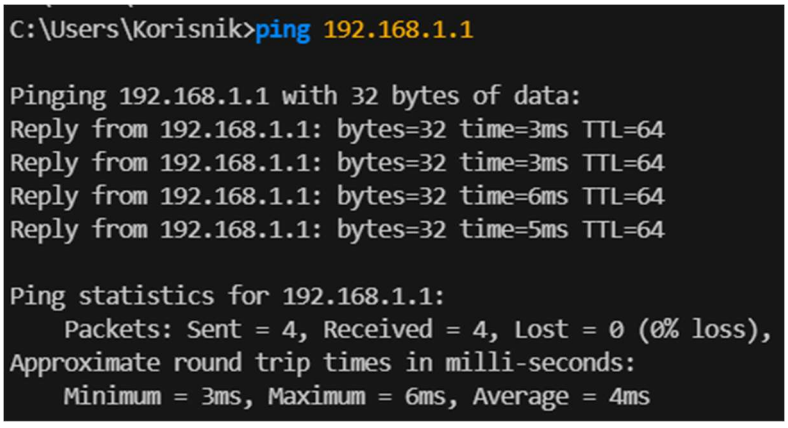
The active network connection on this computer is a wireless Wi-Fi connection using a Realtek 8822CE-VS Wireless LAN 802.11ac PCI-E NIC adapter. The computer is connected to a local home network with the SSID MIWIFI_RrTJ_EXT, as identified in the previous Resource Monitor analysis. Additionally, a Hyper-V Virtual Ethernet Adapter is present as a virtual switch for local VM communication, but this is not a physical network connection and is excluded from the main analysis.

2. **Use ipconfig /all to identify and record:**

- **IPv4 address :** 192.168.1.250
- **Subnet mask.** : 255.255.255.0
- **Default gateway** : 192.168.1.1
- **DNS servers :** 212.230.135.1 and 212.230.135.2
- **DHCP status :** Enabled
- **Network adapter** : Realtek 8822CE-VS Wireless LAN 802.11ac PCI-E NIC

3. **Use the ping command to test connectivity with:**

- **Your default gateway.**



- **A public website or service.**

```
C:\Users\Korisnik>ping google.com

Pinging google.com [172.217.20.238] with 32 bytes of data:
Reply from 172.217.20.238: bytes=32 time=7ms TTL=120
Reply from 172.217.20.238: bytes=32 time=7ms TTL=120
Reply from 172.217.20.238: bytes=32 time=9ms TTL=120
Reply from 172.217.20.238: bytes=32 time=7ms TTL=120

Ping statistics for 172.217.20.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 9ms, Average = 7ms
```

4. Compare the results and explain what successful or unsuccessful responses indicate.

Both pings returned 4 out of 4 packets with 0% loss, confirming full connectivity at both the local network level and the internet level.

The gateway ping averaging 4ms confirms the local Wi-Fi connection to the router is stable and fast. The google.com ping averaging 7ms confirms the internet connection is working correctly and with excellent latency. Anything under 50ms for a public server is considered good for a home connection.

The gateway returned TTL=64 (one hop away on the local network) while Google returned TTL=120 (multiple hops across the internet).

5. Use tracert to examine the route followed by the connection to a public website.

```
C:\Users\Korisnik>tracert google.com

Tracing route to google.com [172.217.20.238]
over a maximum of 30 hops:

  1     3 ms     3 ms     2 ms  HOME [192.168.1.1]
  2    42 ms     7 ms     6 ms  10.96.0.1
  3    24 ms     7 ms     7 ms  10.14.0.113
  4    11 ms     8 ms     7 ms  10.14.246.6
  5     *        *        *     Request timed out.
  6     8 ms     6 ms    12 ms  72.14.194.132
  7    36 ms     7 ms    10 ms  lcmada-ah-in-f14.1e100.net [172.217.20.238]

Trace complete.
```

The route from this computer to google.com passed through 7 hops:

Hop	IP / Host	Avg time	Notes
1	192.168.1.1 (HOME)	3ms	Local router - default gateway
2	10.96.0.1	~18ms	ISP first router
3	10.14.0.113	~13ms	ISP internal network

4	10.14.246.6	~9ms	ISP internal network
5	* * *	Timeout	Router not responding to ICMP
6	72.14.194.132	~9ms	Google network entry point
7	172.217.20.238 (lcmada-ah-in-f14.1e100.net)	~18ms	Google destination server

The route shows a clean path from the local network through the ISP infrastructure and into Google's network in 7 hops. Hops 2–4 use private 10.x.x.x addresses, meaning they are internal ISP routers not exposed to the public internet. Hop 5 timed out. The final destination was reached successfully at an average of 18ms. The hostname lcmada suggests the Google server serving this connection is located in Madrid, which matches the current location.

6. Use nslookup to perform DNS resolution tests for at least three public domain names.

All three queries were handled by the ISP DNS server at 212.230.135.1, which was listed as "UnKnown" for its hostname. All three domains resolved successfully. The responses are marked as "non-authoritative" – this means the answer came from the ISP's DNS cache rather than directly from the domain's own authoritative name server.

```
C:\Users\Korisnik>nslookup google.com
Server:  UnKnown
Address:  212.230.135.1

Non-authoritative answer:
Name:    google.com
Addresses:  2a00:1450:4003:809::200e
          172.217.20.238


C:\Users\Korisnik>nslookup github.com
Server:  UnKnown
Address:  212.230.135.1

Non-authoritative answer:
Name:    github.com
Address:  140.82.121.3


C:\Users\Korisnik>nslookup youtube.com
Server:  UnKnown
Address:  212.230.135.1

Non-authoritative answer:
Name:    youtube.com
Addresses:  2a00:1450:4003:817::200e
          216.58.205.238
```

7. **Use netstat to examine active network connections on the computer and identify at least five active connections.**

```
Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    192.168.1.250:49348    150.171.109.81:443     ESTABLISHED
  TCP    192.168.1.250:49349    150.171.110.152:443    ESTABLISHED
  TCP    192.168.1.250:49350    35.186.224.24:443      ESTABLISHED
  TCP    192.168.1.250:49351    20.42.73.31:443        ESTABLISHED
  TCP    192.168.1.250:49352    31.13.83.51:443        ESTABLISHED
  TCP    192.168.1.250:49674    192.168.1.231:57785    TIME_WAIT
  TCP    192.168.1.250:50928    74.242.255.116:443     ESTABLISHED
  TCP    192.168.1.250:52950    212.230.135.1:53       TIME_WAIT
  TCP    192.168.1.250:53063    99.84.9.96:443         ESTABLISHED
  TCP    192.168.1.250:53329    212.230.135.1:53       TIME_WAIT
  TCP    192.168.1.250:53330    52.110.20.6:443        TIME_WAIT
  TCP    192.168.1.250:53331    52.110.20.6:443        TIME_WAIT
  TCP    192.168.1.250:53332    72.153.5.130:443       ESTABLISHED
  TCP    192.168.1.250:53794    160.79.104.10:443      CLOSE_WAIT
  TCP    192.168.1.250:55356    74.125.206.188:5228    ESTABLISHED
  TCP    192.168.1.250:56847    4.207.247.137:443      ESTABLISHED
  TCP    192.168.1.250:57015    212.230.135.1:53       TIME_WAIT
  TCP    192.168.1.250:57295    31.13.83.51:443        ESTABLISHED
  TCP    192.168.1.250:60262    140.82.113.26:443      ESTABLISHED
  TCP    192.168.1.250:61928    35.186.224.24:443      TIME_WAIT
  TCP    192.168.1.250:61931    20.190.177.84:443      TIME_WAIT
  TCP    192.168.1.250:62167    212.230.135.1:53       TIME_WAIT
  TCP    192.168.1.250:63109    160.79.104.10:443      ESTABLISHED
```

Five active connections examined:

Local address	Foreign address	Port	State	Likely service
192.168.1.250:49351	20.42.73.31:443	HTTPS	ESTABLISHED	Microsoft services
192.168.1.250:49352	31.13.83.51:443	HTTPS	ESTABLISHED	Facebook/Meta
192.168.1.250:53063	99.84.9.96:443	HTTPS	ESTABLISHED	Amazon CloudFront (AWS)
192.168.1.250:55356	74.125.206.188:5228	Push	ESTABLISHED	Google Push Notifications
192.168.1.250:60262	140.82.113.26:443	HTTPS	ESTABLISHED	GitHub

Almost all connections use port 443 (HTTPS), confirming all traffic is encrypted.

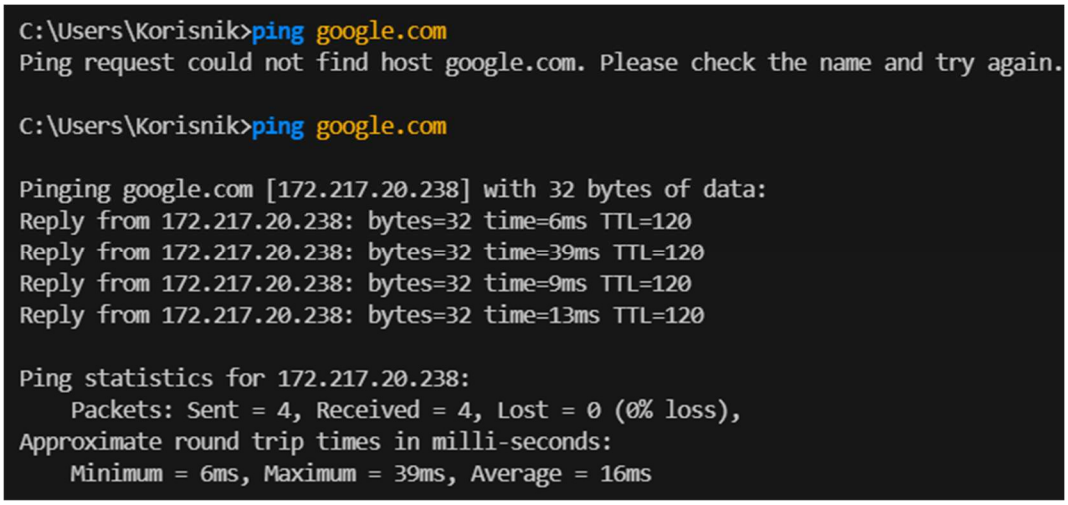
8. **Check the current network adapter configuration through Windows Settings and compare it with the information obtained through Command Prompt.**

	Windows Settings	ipconfig /all (CMD)
IPv4 address	192.168.1.250	192.168.1.250
Default gateway	192.168.1.1	192.168.1.1
DNS servers	212.230.135.1, 212.230.135.2	212.230.135.1, 212.230.135.2
MAC address	00:45:E2:31:DA:25	00-45-E2-31-DA-25
Adapter description	Realtek 8822CE-VS Wireless LAN 802.11ac PCI-E NIC	Realtek 8822CE-VS Wireless LAN 802.11ac PCI-E NIC
IPv6 address	fe80::938c:8c32:53df:4d2	fe80::938c:8c32:53df:4d2%26

Both sources report identical network configuration values, confirming the data is consistent. The only cosmetic difference is the MAC address format. Windows Settings uses colons (00:45:E2:31:DA:25) while ipconfig uses dashes (00-45-E2-31-DA-25), both representing the same physical address.

Windows Settings provides additional information not visible in ipconfig – the SSID, Wi-Fi protocol (802.11ac), security type (WPA2-Personal), channel (5 GHz, channel 100), link speed (433/433 Mbps), driver version, and manufacturer. The Command Prompt provides additional detail about DHCP lease times, DHCPv6 settings, and NetBIOS status.

9. **Disconnect and reconnect the Wi-Fi or Ethernet connection and verify that network connectivity is restored correctly.**



While disconnected:Running ping google.com immediately returned: "Ping request could not find host google.com. Please check the name and try again." This error indicates that DNS resolution failed – with no network connection, the computer could not reach the DNS server.

After reconnecting:The same ping command succeeded immediately – all 4 packets were received with 0% loss and an average response time of 16ms.

10. **Perform a basic troubleshooting sequence for a connectivity problem using the tools analyzed during the task.**

Step 1: Check physical connection

Verify that Wi-Fi is enabled and connected to the correct network. In this task, disconnecting Wi-Fi immediately caused DNS resolution to fail, confirming that no software tool can help if the physical or wireless connection is down.

Step 2: Run ipconfig /all

Check that the adapter has a valid IP address, subnet mask, default gateway, and DNS servers. If the IP address starts with 169.254.x.x it means DHCP failed and the computer assigned itself an address, indicating the router is unreachable or DHCP service is down.

Step 3: Ping the default gateway

Ping the default gateway. If this fails, the problem is between the computer and the router. The issue is local, wrong Wi-Fi password, router offline, or faulty cable.

Step 4: Ping a public IP address

Ping 8.8.8.8 for example, because it bypasses DNS and tests raw internet connectivity. If this succeeds but domain names don't resolve, the problem is DNS-specific, not connectivity.

Step 5: Run nslookup

nslookup google.com – tests whether the DNS server is responding and resolving names correctly. If this fails, the DNS server is unreachable or misconfigured.

Step 6: Run tracert

tracert google.com - if connectivity is partial, tracert identifies exactly where the route breaks. The last responding hop before timeouts points to where the problem is located.

Step 7: Run netstat

netstat -n – if connectivity seems fine but a specific application is not working, netstat shows whether the application has an active connection and what state it is in.

11. **Create a table containing: Command used, Purpose, Result obtained, Interpretation of the result.**

Command	Purpose	Result obtained	Interpretation
ipconfig /all	Display full network adapter configuration	IP 192.168.1.250, gateway 192.168.1.1, DNS 212.230.135.1 and 212.230.135.2, DHCP enabled	Network is correctly configured via DHCP with ISP-provided DNS servers
ping 192.168.1.1	Test connectivity to the local router	4/4 packets received, 0% loss	Local network connection to the router is stable and fast
ping google.com	Test internet connectivity and DNS resolution	4/4 packets received, 0% loss	Internet connection is working with excellent latency
tracert google.com	Trace the route to a public server	7 hops to Google, destination reached at ~18ms	Clean route through ISP to Google's Madrid server
nslookup google.com	Test DNS resolution for a domain	Resolved to 172.217.20.238 via DNS server 212.230.135.1	DNS is working correctly, ISP DNS server responding normally
nslookup github.com	Test DNS resolution for a second domain	Resolved to 140.82.121.3	DNS resolution working for multiple domains
nslookup youtube.com	Test DNS resolution for a third domain	Resolved to 216.58.205.238	DNS resolution working, YouTube returning both IPv4 and IPv6 addresses
netstat -n	Display active network connections	23 connections identified, mostly ESTABLISHED on port 443	All active traffic is encrypted HTTPS, multiple services running in background